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The Matachewan dyke swarm, Canada: an early Proterozoic magnetic field reversal

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ABSTRACT

Consistent age relationships between oppositely magnetized dykes of the 2.45 Ga Matachewan dyke swarm suggest that only a single magnetic field reversal occurred during the period of igneous activity. The magnetic field throughout most of this time was characterized by a SSW declination and shallow negative inclination but reversed toward the waning stages of magmatism. The new paleomagnetic data provide the oldest known magnetic reversal for which the relative reversal sense is known.

1. Introduction

In this paper I report paleomagnetic data that bear on the reversal history of the earth's magnetic field during the period of igneous activity when the early Proterozoic Matachewan dyke swarm was emplaced over about 250,000 km^2 of the Canadian Shield [1].

The Matachewan swarm, comprising many hundreds to thousands of dykes, fans northwards from a broad focus in the Lake Huron region (inset, Fig. 1). The median dyke width is about 20 m and the total amount of crustal extension across the full 400 km of the swarm is about 1–2%. The total volume of magma is about 100,000 km^3. The dykes are iron-rich tholeiites often characterized by pale green plagioclase phenocrysts distributed in layers parallel to the dyke margin. The radiating pattern of the swarm is distorted on crossing the Kapuskasing Structural Zone (KSZ, Fig. 1) where deep crustal rocks of amphibolite to granulite facies have been upthrust to the surface [2]. The dykes have experienced at most a lower greenschist facies of metamorphism and exhibit varying degrees of hydrous alteration that bear an inverse relation to the regional metamorphic grade of the host rocks, reaching a maximum (ubiquitous chlorite, sericite and fibrous amphibole) south and west of the KSZ and a minimum within the KSZ [3].

Precise radiometric dating of the Matachewan swarm, using the U-Pb method on zircons and baddeleyite from three dykes yields an age of $2452 + 3/-2$ Ma [4,5]. Two of the dykes have been paleomagnetically studied [3, sites 208 and 214] and found to be of opposite polarity. The data suggest that the Matachewan swarm was emplaced in a geologically short period of time of less than 5 Ma [5].

Paleomagnetic data, based on more than 2000 oriented samples from about 300 dykes, show that the Matachewan swarm has been regionally deformed, through differential crustal uplift and block rotation associated with the development of the Kapuskasing Zone about 2 Ga ago [1,3,6–8]. These results show that the distorted appearance of the swarm is a secondary feature and that it originally had a more linear radiating pattern.

Remanent magnetization directions of opposite polarity have been known in Matachewan dykes since they were first studied more than 25 years ago [9–11]. The remanence polarity with respect to the present earth's field is unknown because the apparent polar wander path for North America is

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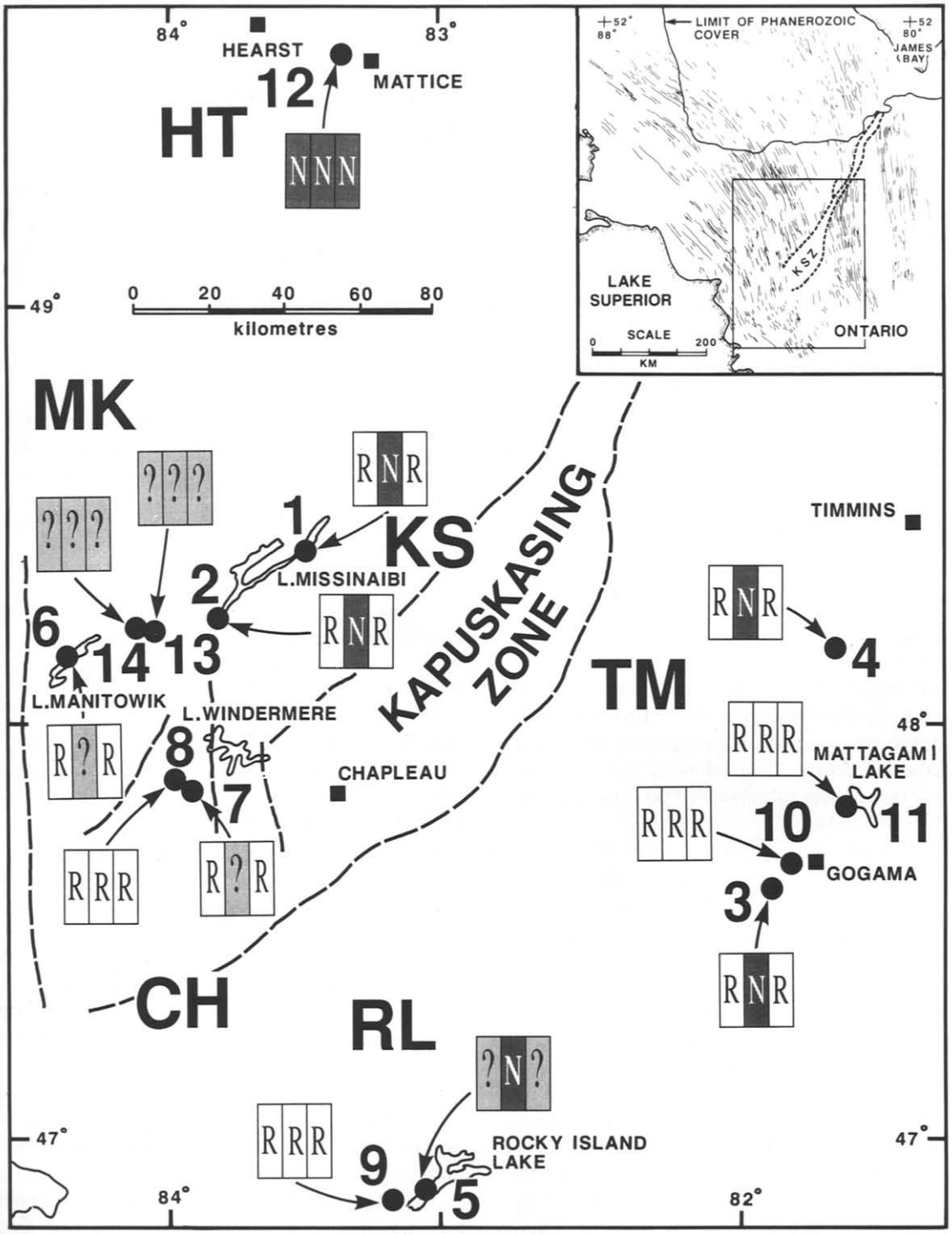


Fig. 1. Map showing Locations where age–polarity relations in Matachewan dykes were studied. All of the locations comprise pairs of intersecting dykes except Locations 3 and 4 where parallel dykes of normal and reversed magnetic polarity lie in close proximity. The large letters *HT*, *MK* etc. refer to the prefix label for all paleomagnetic sites collected in the vicinity of where the letters are placed. Flags schematically portray a dyke cutting and splitting an older one and they are shaded according to the polarity relations found at each locality. The paleomagnetic sites corresponding to the fourteen locations are: Location *1* = KS4,21; *2* = KS11,19; *3* = TM43,45; *4* = TM5,6; *5* = RL8; *6* = MK8; *7* = CH209; *8* = CH205; *9* = RL23; *10* = TM29; *11* = TM12; *12* = HT12; *13* = KS14; *14* = MK2. The inset map shows the general configuration of the Matachewan swarm in northern Ontario and the location of the study area at the southern end of the Kapuskasing Structural Zone.

ill-defined for the Precambrian [12]. However for convenience the direction closest to the present field will be referred to as the "normal" one. For the Matachewan dykes this "normal" (N) polarity direction has a shallow positive inclination and NNE declination and the "reversed" (R) polarity remanence is virtually antipodal to N. Two lines of evidence suggest that these remanences are primary, either acquired thermally and/or chemically during cooling of the dykes: dykes of opposite polarity may cut or occur within a few metres of one another [3] and both R and N dykes have positive baked contact tests [13,14].

The Matachewan swarm is dominated by R dykes which outnumber those of N polarity by about 4:1 [1,3,8]. However, two regions, one within the KSZ around Chapleau, the other 10 km west of Windermere Lake (Fig. 1) are characterized by N polarity dykes [3]. Both these areas are bordered on their east and southeast sides by major faults across which the dyke population changes to R polarity [3,8]. Cloudy feldspars in dykes which are thought to be diagnostic of crystallization at deep crustal levels [3] occur in both N domains but not in the R ones. The N domains thus represent relatively uplifted crustal segments and show that deformation related to the KSZ occupied a broad area. Critical to the understanding of the polarity discontinuities and to the evolution of the Matachewan swarm are the age relations between N and R dykes [3].

2. Paleomagnetic investigations

2.1. Experimental procedure

The primary nature of the N and R remanences indicates that dyke intrusion spanned one or more reversals of the earth's magnetic field. The case for a single reversal can be made only if it can be shown that dykes of one polarity always bear the same relative age to dykes of the other polarity. In this instance the polarity of the dyke would give an immediate indicator of whether the dyke was emplaced early or late in the evolution of the swarm. In this paper I test for a consistent polarity relationship through detailed geological and paleomagnetic studies at fourteen locations (Fig. 1) where Matachewan dykes either intersect or lie adjacent to one another. This study is the outcome of seven field seasons which were needed to discover and document the relatively rare dyke intersections, and to return to selected sites once preliminary paleomagnetic data indicated that Matachewan remanences of opposite polarity were present.

Paleomagnetic samples were collected either in blocks or as field-drilled cores, whose orientations were recorded twice using a magnetic and solar compass. Their remanent magnetism was measured using a modified Digico magnetometer, and detailed alternating field demagnetization to 100 mT was performed using a Schonstedt SD-1 demagnetizer. For each sample, demagnetization steps were tailored so that a smooth demagnetization path was produced. For each field above 25 mT data were generally averaged to reduce directional scattering, which is a feature of single axis demagnetizers once intensities fall below about 20% of the initial intensity. Data were processed and analysed using a procedure [15] involving principal component analysis [16] with accepted component directions having maximum angular deviation values $\leq 10°$.

2.2. Paleomagnetic results

A summary of the paleomagnetic results for all locations in Fig. 1 where Matachewan directions were recorded is given in the Appendix.

Locations 1 and 2 are about 30 km apart, at the opposite ends of Lake Missinaibi (Fig. 1) and in each case an older dyke more than 20 m thick is obliquely cut by dykelets up to several centimetres wide. All dykes display excellent chilled margins against their host rocks. Location 1 comprises two paleomagnetic sites on the opposite sides of a small island (Fig. 2). Paleomagnetic data are compared for each of three dykes that can be correlated across the island (Fig. 3). At the first site (KS4) a thick, weakly porphyritic R dyke occurred to the east of two thinner non-porphyritic N dykes (Figs. 2 and 3A–C). At site KS21 two N dykes were identified (Fig. 3D, E), the eastern one of which was observed to cut an older R dyke (Fig. 3F). The older dyke could be followed discontinuously across the island and equated with the R dyke at site KS4. A summary of paleomagnetic directions from Location 1 (Fig. 4) shows a 10° asymmetry in declination between the N and R components.

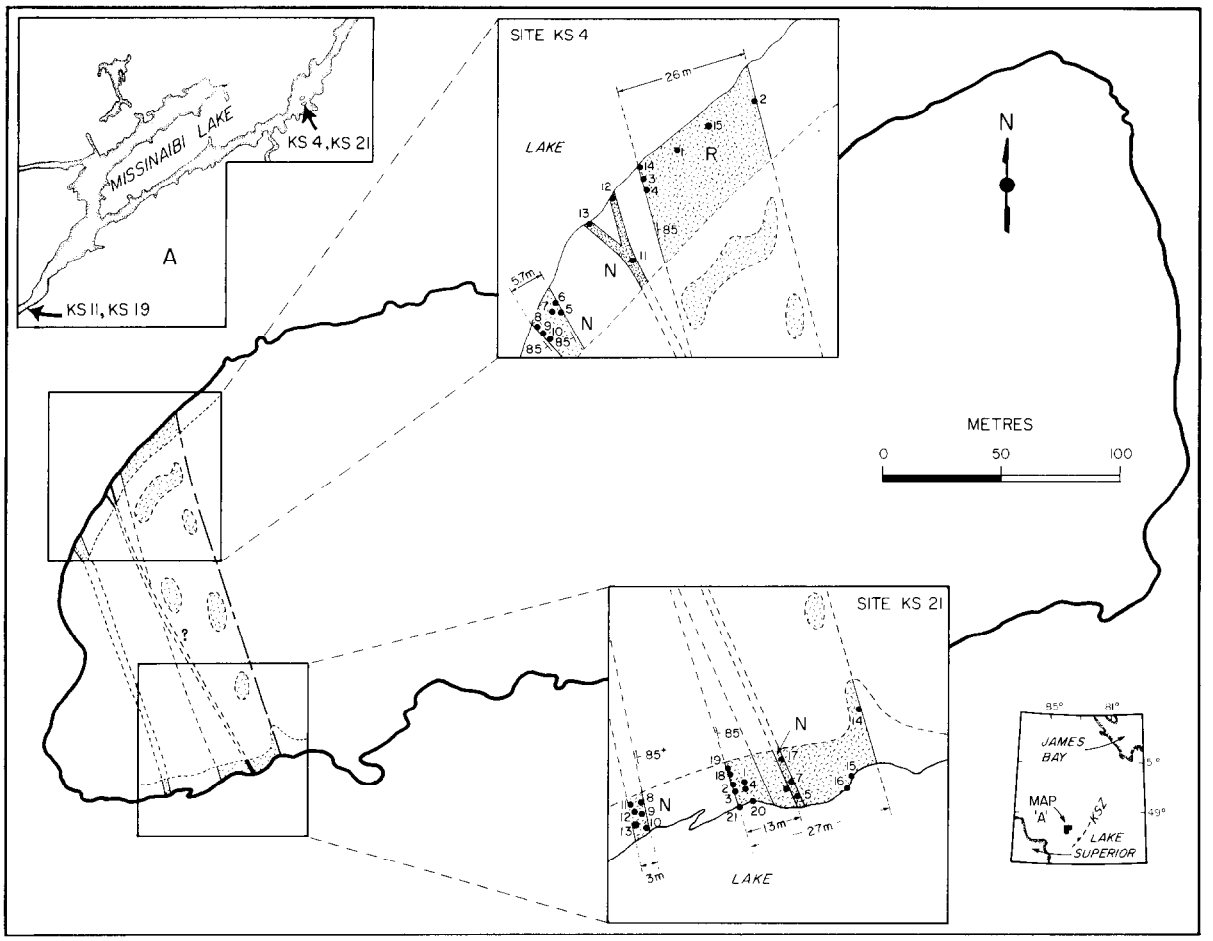


Fig. 2. Geological map of Location 1, comprising paleomagnetic sites KS4 and KS21. Diabase dykes are shown in the stippled pattern, where *N* and *R* denote a normal or reversed magnetic polarity dyke. Locations of paleomagnetic samples are indicated by black dots. Samples KS21-14 to 21 were field drilled, the remainder were block sampled. At site KS21 a second, unsampled dykelet is shown cutting the *R* dyke.

At Location 2 (Fig. 5) the older dyke is locally porphyritic in its central and marginal parts and is cut by three non-porphyritic dykelets. Both dyke generations are cut by a NE-trending system of pale green to white veinlets. The younger dykes are of *N* polarity (Fig. 6A, B) and the older dyke is *R* (Fig. 6C–F). However, many of the samples are severely overprinted with a remanence component with a steep upward direction of magnetization (Fig. 7) that is associated with an extensive NE-trending zone of dyke intrusion [3] and shearing that occurs along Lake Missinaibi. This component is found in both *N* and *R* dykes (Fig. 6B–E) and is probably a CRM associated with the veining. At locations 13 and 14, less than 20 km to the west, crosscutting Matachewan dykes are completely overprinted by a similar steep upward component. At location 14 both dykes are cut by E–W sinistral shear zones with mineralized tension gashes. The directions of Matachewan components obtained from Location 2 (Fig. 7) show a pronounced asymmetry between the *N* and *R* components, a possible consequence of incomplete removal of the magnetic overprint.

At locations 3 and 4, dykes of opposite polarity occur in proximity to one another. At location 3 (inset, Fig. 8) a 27 m wide dyke is entirely of *R* polarity (including host rock near the east margin)

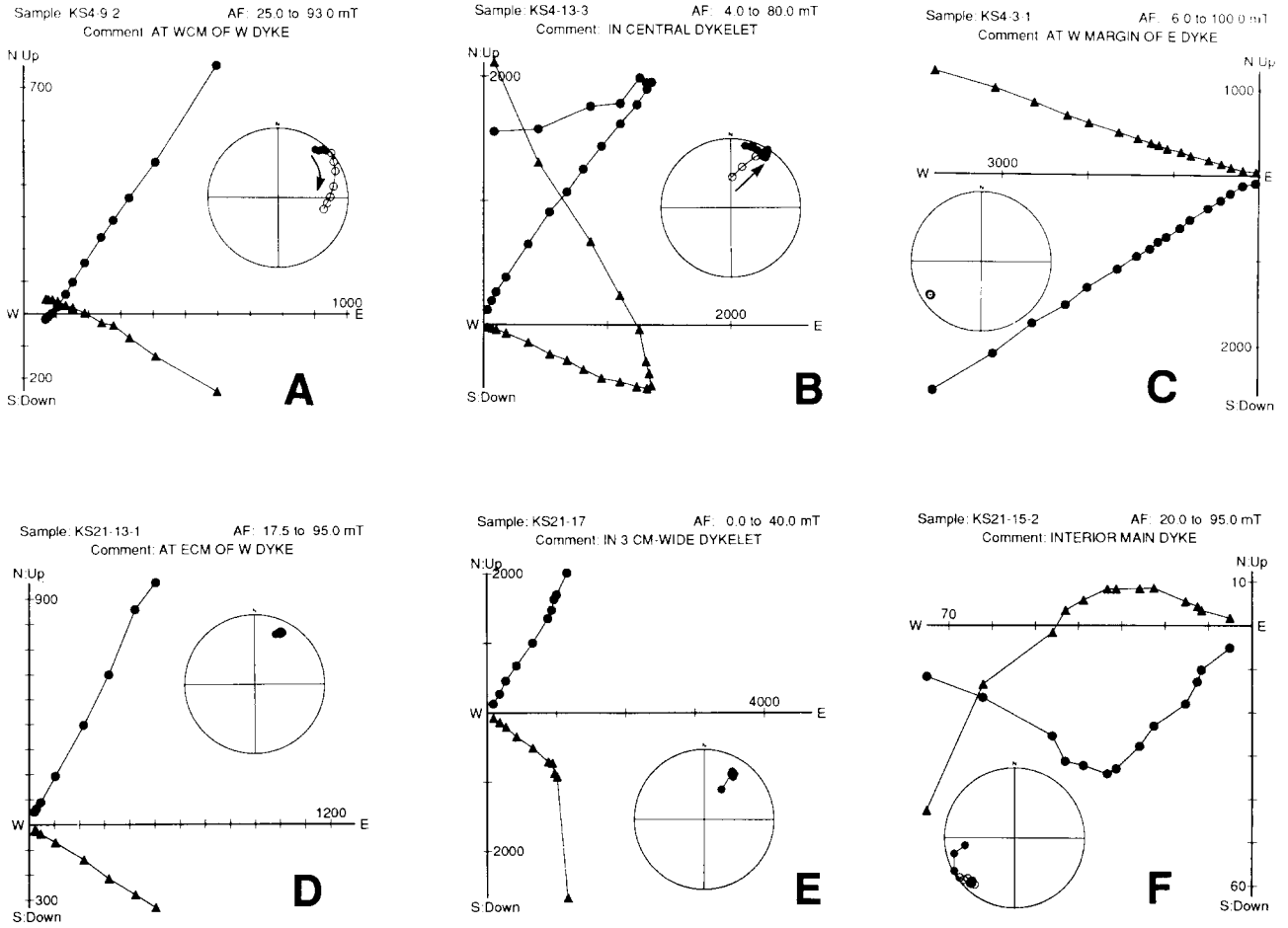


Fig. 3. Paleomagnetic data from Location 1. Equal area stereoplots and associated orthogonal vector diagrams of a representative sample from each dyke from site KS4 (A–C) and site KS21 (D–F). Arrangement of diagrams is approximately geographical (north up and west to the left). In vector diagrams circles are projections of the tip of the remanence vector onto the horizontal plane; triangles are projections onto an E–W vertical plane. Units are Amperes/metre $\times 10^{-3}$. In stereoplots open/closed circles refer to upward/downward-pointing magnetizations.

but within several decimetres of the western margin a single component *N* behaviour is found (Fig. 9A–C). A careful re-check of the dyke shows it to be a single intrusion. About 8 m to the west is a 6 m wide dyke which is entirely *N* (Fig. 9D) including host rock samples along its eastern margin.

Detailed paleomagnetic studies of dykes and their remagnetization effects on adjacent crystalline rocks of the Canadian Shield have been carried out in several places [17–19]. In all cases the hybrid zone of partial resetting in the host rocks was observed at distances from the dyke of about one third to one half its width. Therefore, at location 3 the *N* dyke is the younger, otherwise it would have been completely reset by the thick *R* dyke. The *N* remanence within the *R* dyke is thus interpreted as a TRM/CRM associated with the intrusion of the younger dyke. Although sampling is less, a virtually identical situation occurs at Location 4 (inset, Fig. 8 and Fig. 9E–H).

Of the ten remaining locations, all involving pairs of crosscutting dykes, three (Nos. 8, 9 and 11) involve only *R* dykes, one (No. 12) only *N* dykes, two (Nos. 13 and 14) are totally overprinted and the other four (Nos. 5, 6, 7 and 10) are incomplete as one dyke is either magnetically

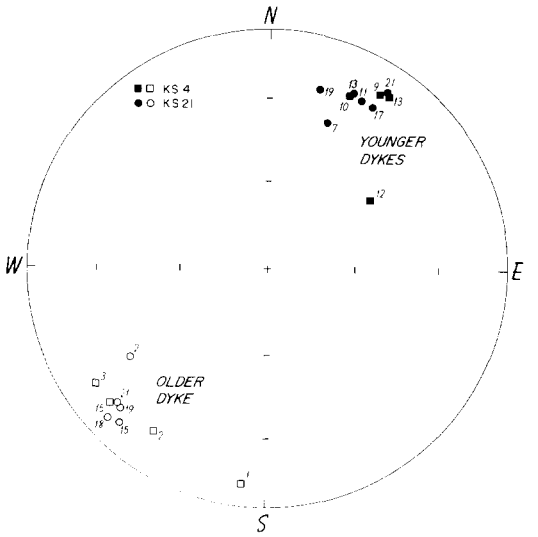


Fig. 4. Equal area stereonet summarising interpreted directions of primary remanence from Location 1. Open (closed) symbols refer to upward (downward) pointing magnetization directions. Numbers beside points refer to sample locations in Fig. 2.

unstable or has a remanence that represents a total resetting by the other dyke. However, none of these inconclusive sites yields data in conflict with previous observations that N is younger than R. However, one of them (No. 10) provides indirect evidence of the polarity sequence, and is discussed below after a brief description of the general demagnetization characteristics of Matachewan dykes.

For both N and R remanences median destructive fields are less than 10 mT, typical initial intensities are of the order of 1 A/m and fall rapidly so that by demagnetization fields of 20 mT less than 20% of the initial intensity (J_0) remains. At low coercivities the Matachewan components are completely masked by viscous and other components that may be related to younger geological events. Only after about 20 to 40 mT are these components removed, whereupon the remanence remaining may fall below 10% J_0 so that linear

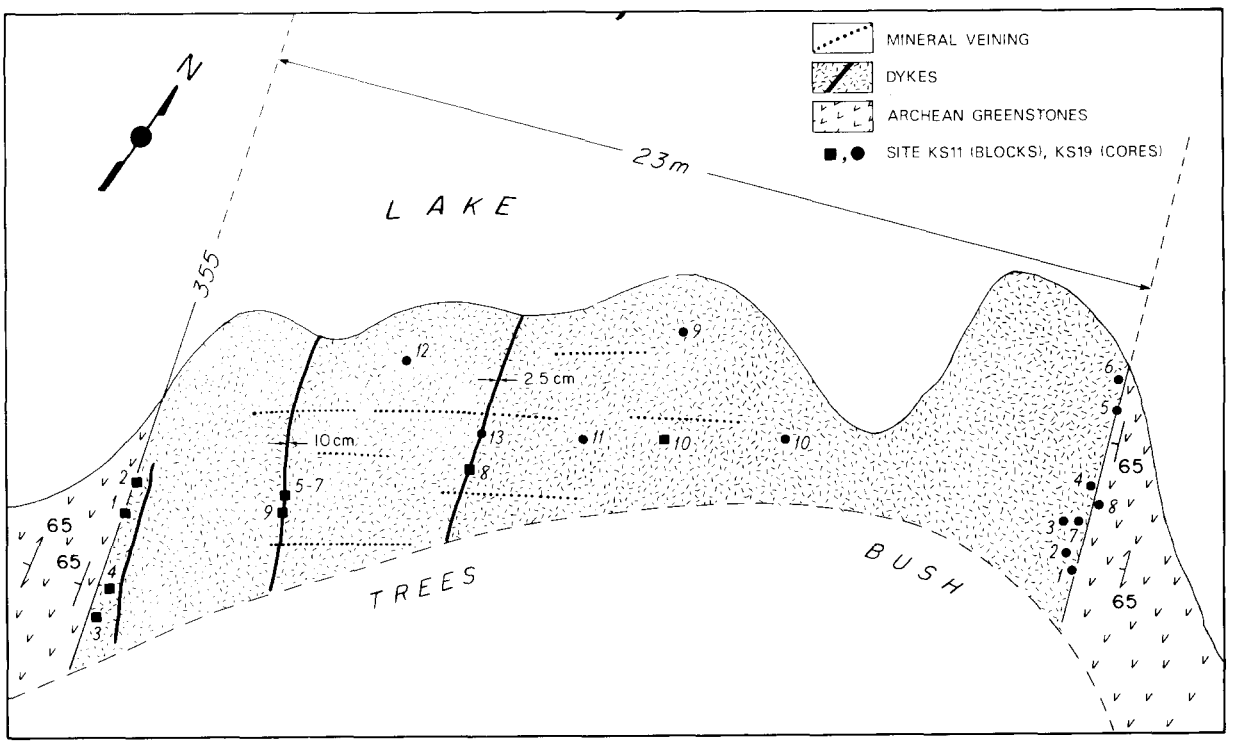


Fig. 5. Geological sketch map showing the distribution of paleomagnetic samples at Location 2 comprising paleomagnetic sites KS11 and KS19. See inset of Fig. 2 for location. Both generations of dykes are cut by a system of narrow (less than a few mm) NE-trending epidote (?) veinlets.

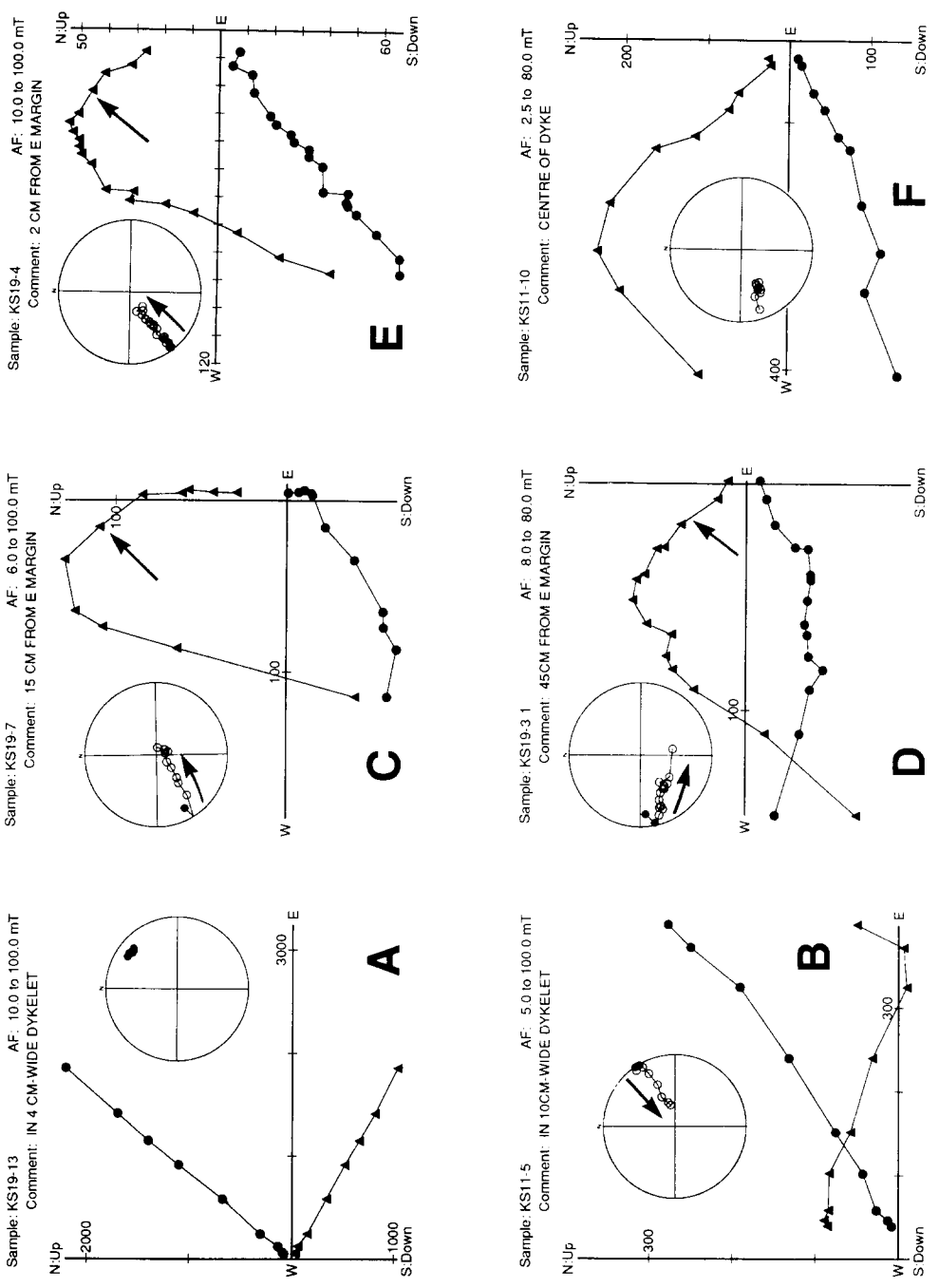


Fig. 6. Orthogonal vector plots from Location 2. *N* polarity samples from the younger dykes are either single component (A) or show a second component with a steep upward magnetization (B) which is well displayed in samples from the older dyke (C–E) and which may be related to the mineral veining. The older dyke carries an *R* polarity remanence which appears mostly as an intermediate component (arrowed in diagrams C–E) or as a stable end point (F). Symbols the same as in Fig. 3.

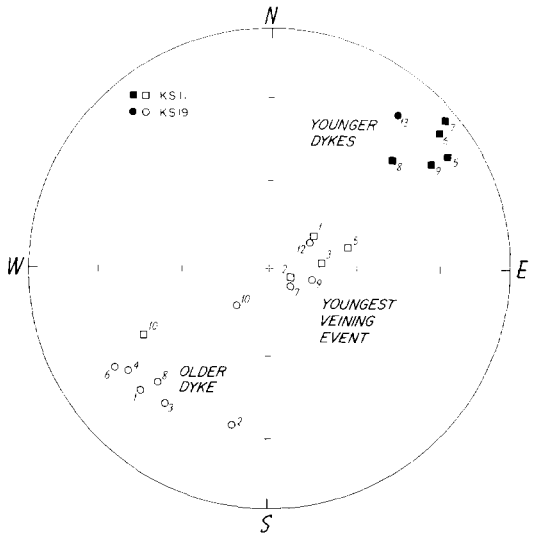


Fig. 7. Equal area stereonet summarizing the magnetization components obtained at Location 2. Open/closed symbols refer to upward/downward directions of magnetization. Numbers beside points refer to sample locations in Fig. 5.

segments on vector diagrams are ill-defined without careful demagnetization procedures. Of the 2000 or so Matachewan samples that have been demagnetized, less than 1% show the unequivocal presence of *both* N and R components. These samples show three kinds of demagnetization behaviour (Fig. 10): (a) samples that carry a strong high coercivity R component have a small low coercivity N that is removed by about 10 mT (Fig. 10A); (b) Some dykes show normal intensity R interiors (Fig. 10A, left) but have chilled margins of anomalously low intensity that contain an R component with a smaller higher coercivity N (Fig. 10B, right); and (c), the most common observation, is that normal intensity R dykes have chilled margins that contain a small high-coercivity N component that can be isolated only at fields above 50 mT (Fig. 10C). Noteworthy in all three instances is the complete lack of the converse polarity relationship.

An important observation is that the same behaviour depicted in Fig. 10C is also found in the chilled margin of the R dyke at Location 1 (Figs. 2 and 10D) where younger N intrusions are documented on the basis of crosscutting relations. In this case, the chilled margin is many dyke widths

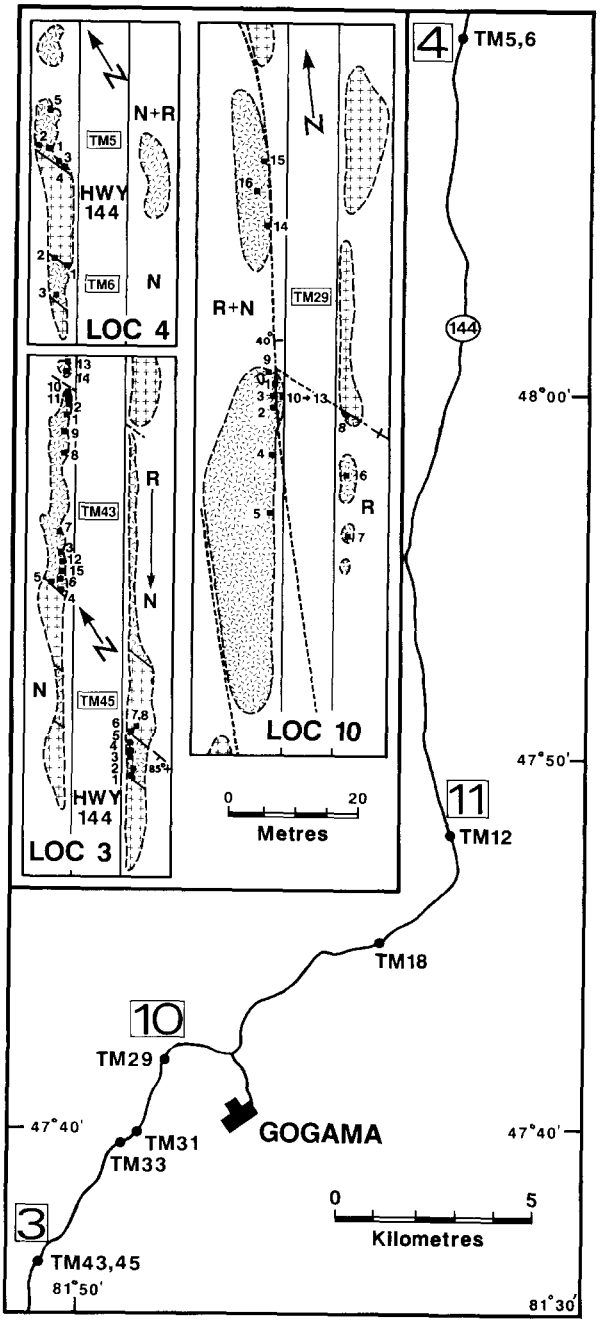


Fig. 8. Map showing four locations (Nos. 3, 4, 10 and 11) where polarity relations can be deduced. Other paleomagnetic sites referred to in the text (TM 18, 31 and 33) are also shown. The inset map shows detailed geological maps of three locations where normal and reversed Matachewan magnetizations are found. Solid squares mark where block samples were collected; numbers beside squares give the sample number. Dykes shown in short line pattern; granite or gneissic host rocks shown with plus signs.

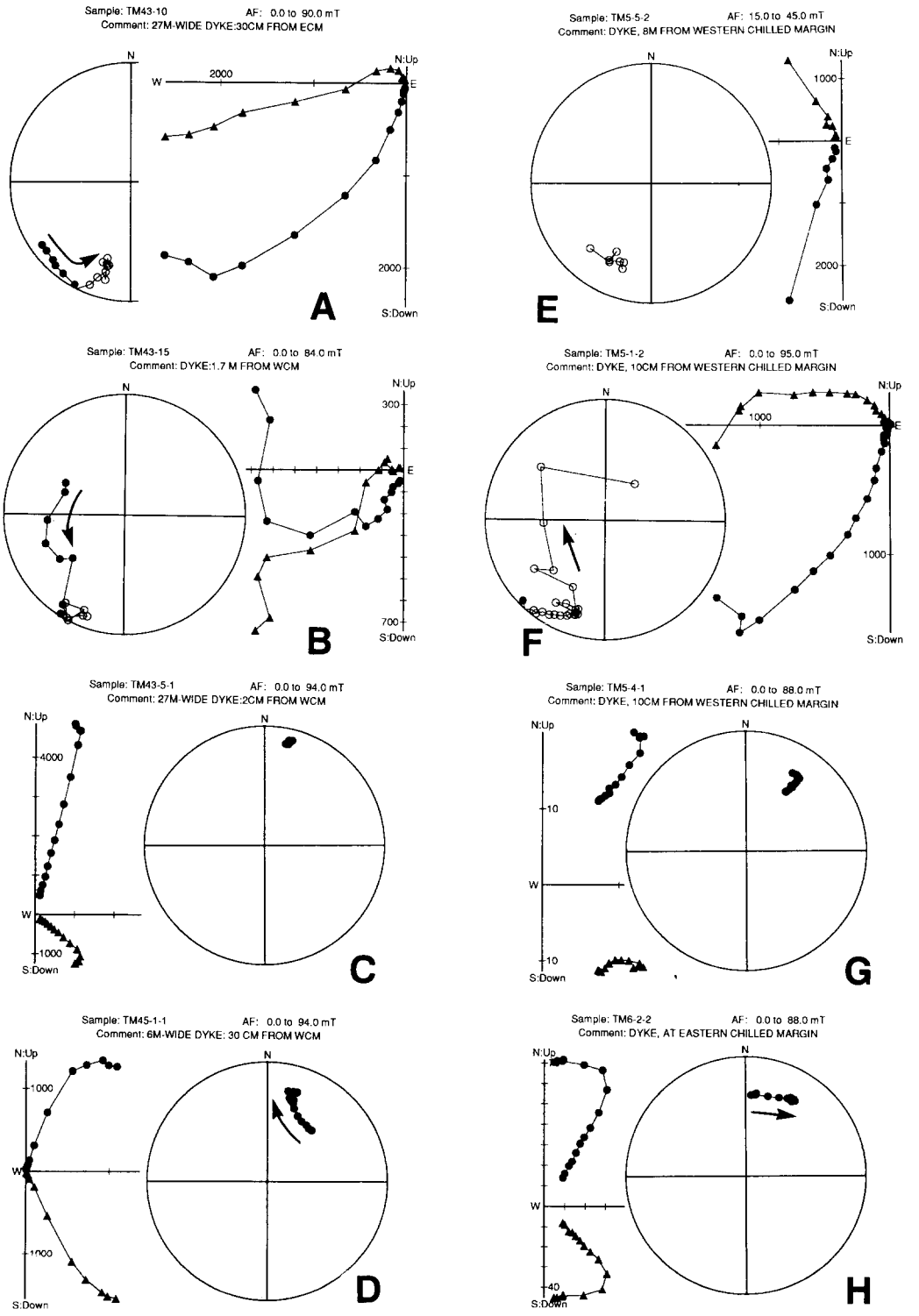


Fig. 9. Equal area stereonets and associated vector diagrams for Location 3 (A–D) and Location 4 (E–H). For each location diagrams are arranged down the page in geographic order from northeast to southwest. Symbols are the same as in Fig. 3.

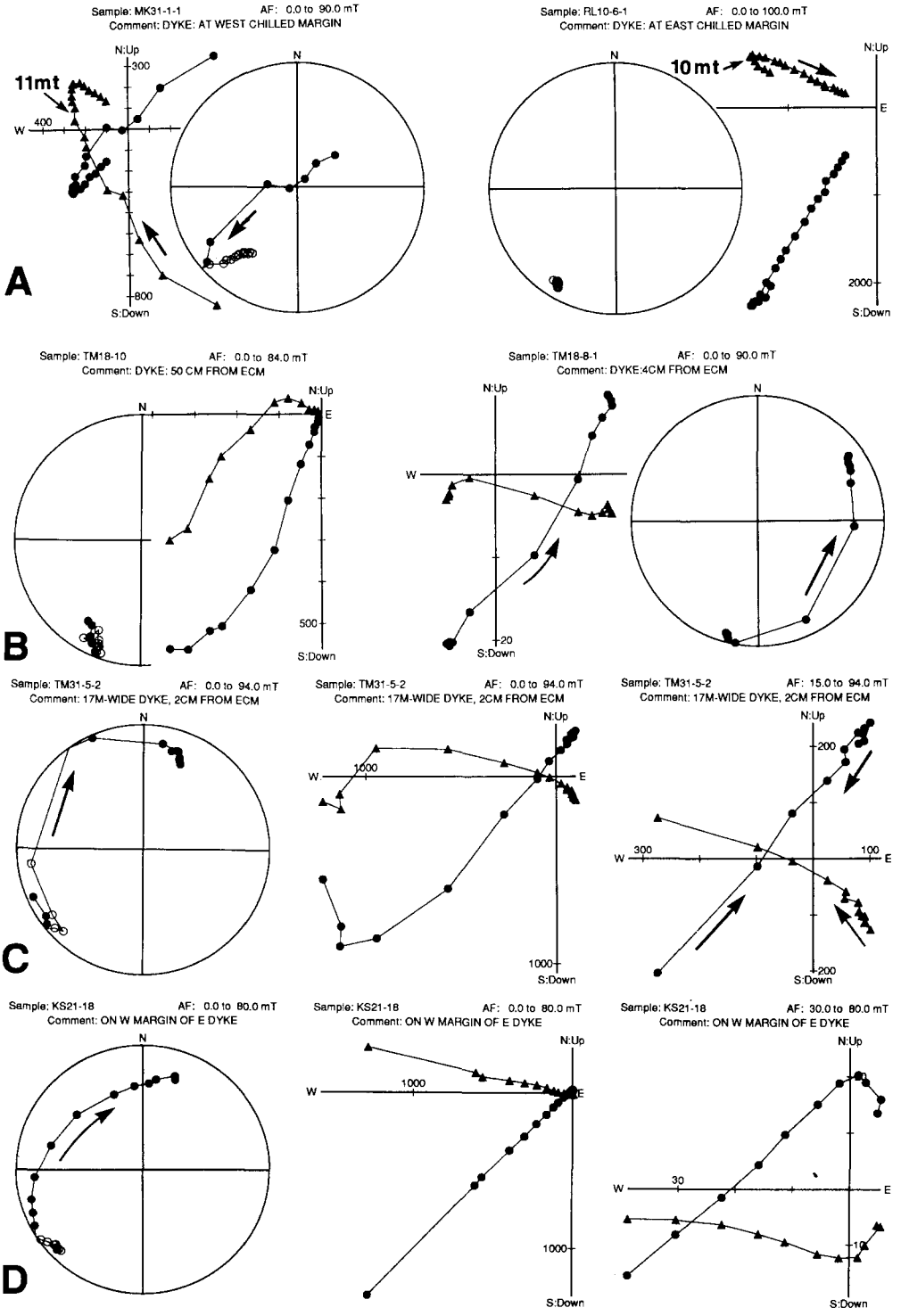


Fig. 10. Equal area stereonets and associated vector diagrams showing three types of behaviour of Matachewan dyke samples containing both N and R remanence components. (A) Low-coercivity N in chilled margins of two R dykes. (B) Typical interior R dyke (left diagram) with anomalously weak intensity $R + N$ chilled margin from the same dyke (right). (C) Chilled margin samples carrying dominant R component plus a weak high-coercivity N. The NRM intensity of the sample is similar to more interior samples which are all R. In diagram D the same behaviour as in C is shown except that it is from Location 1 where the N component is known to be younger than R.

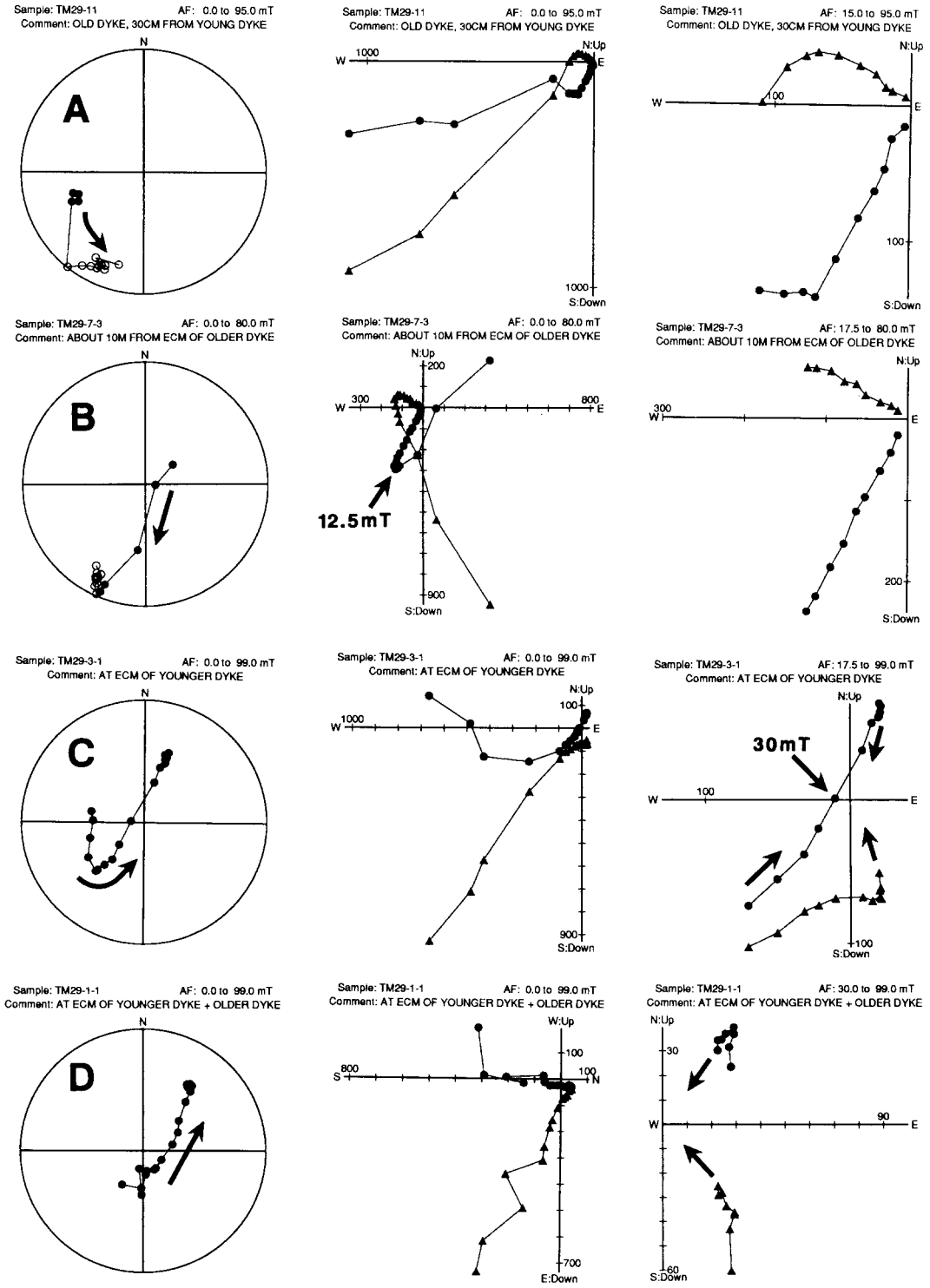


Fig. 11. Equal area and associated vector diagrams from samples at Location 10 (see inset map of Fig. 8 for the geological map). Note that both dykes have *R* polarity. The older dyke is completely reset to *R* by the younger one, at least within a few cm of the younger dyke (A). At greater distances *R* completely dominates (B) but it is less clear if this is a resetting or a primary direction for the older dyke. At the chilled margin of the younger dyke (C, D) a lower coercivity *R* component is replaced by *N* at alternating fields above 50 mT. Since the younger dyke is *R* the *N* component is younger and probably a CRM caused by *N* dyke-related fluids.

away from the nearest N dyke so that the N component is probably a CRM. Chilled margins may be particularly susceptible to resetting if they are fractured and especially if small grains near the superparamagnetic threshold change size as a result of chemical action. Based on the observation at Location 1, small high-coercivity N components in chilled margins of otherwise R dykes are consistent therefore with the younger age of the N component. This behaviour has been observed at Location 10 in the younger dyke (Figs. 8 and 11). This dyke was injected during an R epoch because only 30 cm from it, in the older dyke, a stable R component is observed. A similar observation of a small hard N along one or both margins of otherwise R dykes is observed at single dyke sites TM18, -31 and -33 in Fig. 8. Based on the observations at Location 1 I interpret the N component at all these sites to be a CRM introduced by fluids associated with later N dyke intrusion.

3. Discussion

Paleomagnetic results have been described from fourteen Matachewan dyke locations which offered prospects of gaining information on polarity–age relationships. These relations are summarised in the schematic flags shown in Fig. 1. Four of the locations (Nos. 1–4) provide firm evidence that the N polarity is younger than R and one (No. 10) gives indirectly the same result. It is important to note that none of the remaining locations yield data that are inconsistent with a younger age of the N remanence. Therefore, the data suggest that only one reversal, from $R \rightarrow N$ polarity occurred during the emplacement of the Matachewan dyke swarm. This conclusion is supported by the high-precision U-Pb data in which both the N and R dykes give ages that are not significantly different. Furthermore south of the KSZ in the Abitibi Subprovince, mean paleomagnetic directions obtained from different regions across the swarm are within about 5° of one another [3,8,20] and this similarity supports the conclusion from the U-Pb data that igneous activity lasted only a few million years. The rarity of N dykes compared to R further suggests either that Matachewan igneous activity was waning by the time of the normal epoch, or that it had moved to deeper crustal levels so that abundant N dykes are only exposed in uplifted windows of crust such as those along the KSZ [3]. An alternative explanation for the high frequency of N dykes in these uplifted terrains is that they were intruded during the R epoch but only acquired their remanence after delayed cooling at deeper crustal levels when the magnetic field had reversed.

4. Conclusions

A single reversal of the Earth's magnetic field occurred 2452 Ma ago during the period of Matachewan igneous activity and represents the oldest known reversal for which the relative reversal sense can be established. The reversal thus provides a simple chronometer for the order of dyke emplacement, an important ingredient in establishing how the Matachewan swarm developed.

Acknowledgements

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Appendix

Loc.	Site	UTMG	Older dyke					Younger dyke				
			N	*D*	*I*	*P*	α_{95}	*N*	*D*	*I*	*P*	α_{95}
	KS4											
1		17150670	*7*	*221*	*−15*	*R*	*12*	*8*	*28*	*22*	*N*	*7*
	KS21											
	KS11 [1]											
2		17897494	*6*	*223*	*−31*	*R*	*12*	*6*	*51*	*19*	*N*	*9*
	KS19 [1]											
	TM43		*7*	*210*	*−21*	*R*	*13*					
3		17351728	3	18	16	N	9 [2]					
	TM45								*6*	*18*	*19*	*N 6*
	TM5											
4		17562338	*3*	*201*	*−16*	*R*	*18*					
	TM6							*2*	*28*	*26*	*N*	*4*
5	RL8	17403967	1	138	−69	?	-	*2*	*22*	*−4*	*N*	*24*
6	MK8	16932316	4	201	−22	R	20 [3]	5	50	−48	?	50 [5]
7	CH209	17767077	6	221	−24	R	14 [3]	7	276	69	?	22 [4]
8	CH205	16237082	*6*	*238*	*−33*	*R*	*9*	*1*	*226*	*−36*	*R*	-
9	RL23	17225903	*3*	*199*	*−3*	*R*	*13*	*3*	*203*	*−10*	*R*	*13*
10	TM29	17417828	3	200	−15	R	18 [3]	*5*	*205*	*−8*	*R*	*12*
								3	28	27	N	19 [2]
11	TM12	17559940	*2*	*214*	*−14*	*R*	*14*	*2*	*214*	*−17*	*R*	*18*
12	HT12	13146048	*3*	*18*	*20*	*N*	*12*	*2*	*20*	*27*	*N*	*20*

Numbers in italics indicate direction estimates of remanences that are considered to be primary and thus related to the time of crystallization of the dyke in which they are found.

UTMG = Universal Transverse Mercator Grid, giving locations of sites to within 100 m for all sites except HT12 which is located to within 300 m; *N* = Number of independently oriented samples; *D*, *I* = Mean declination, inclination; *P* = Polarity; and α_{95} = Radius of 95% confidence circle about the mean direction.

[1] Due to possible overlap in coercivity spectra the *N* and *R* directions may be biassed towards a strong overprint ($D = 90$, $I = -75$) which is present in most samples. It completely overprints both dykes at Location 13 (Fig. 1) and may be related to a period of NE dyke intrusion and faulting about 2 Ga ago.

[2] The remanences are interpreted to be secondary, caused by thermal/chemical activity during intrusion of *N* dykes.

[3] This remanence could be secondary due to heating by the younger dyke.

[4] The remanence has a low coercivity and is probably secondary.

[5] Several brecciated shear zones cut through the dyke which is altered and paleomagnetically unstable.

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